

YANG LONG

Department of Biostatistics, Yale School of Public Health

✉ ylong5@gmu.edu [github.io](https://github.com/ylong-chn)

ACADEMIC POSITIONS

Postdoctoral Associate, Department of Biostatistics, Yale University, *New Haven, CT* Aug 2026 – Present
• **Supervisor:** Dr. Yize Zhao

EDUCATION

PhD in Statistical Science, George Mason University, *Fairfax, VA* Jan 2021 – May 2026
• **Advisors:** Dr. David Kepplinger and Dr. Lily Wang
• **Dissertation:** Trustworthy AI Through Robust Functional Data Analysis and Statistical Inference for Imaging Data
MS in Statistics, CUNY–Baruch College, *New York, NY* Jan 2017 – May 2019
MS in Finance, University of Rochester, *Rochester, NY* Jul 2014 – Dec 2015
BEcon in Finance, Zhongnan University of Economics and Law, *Wuhan, China* Sep 2010 – Jun 2014

RESEARCH INTERESTS

- Trustworthy AI
- Robust nonparametric statistics
- Functional data analysis
- Multispectral imaging analysis
- Non-convex optimization
- Time series analysis

PUBLICATIONS

Publications (Published & Accepted)

- P1. **Y. Long**, G. Cao, D. Kepplinger, and L. Wang, "**Robust mean signal estimation and inference for imaging data**," *Statistica Sinica*, Forthcoming, 2026+.
- P2. Z. Li, S. Bruce, C.J. Wutzke, and **Y. Long**, "**Conditional adaptive Bayesian spectral analysis of replicated multivariate time series**," *Statistics in Medicine*, vol. 40, pp. 1989–2005, 2021.
- P3. D. Feldman, S. Gross, and **Y. Long**, "**Gender competitiveness and predictability, and prize money in Grand Slam tennis tournaments**," *Quarterly Journal of Finance*, vol. 10, no. 2, 2020.

Working Papers (Submitted & In Preparation)

- W1. G. Wang,[†] **Y. Long**,[†] S. Yu, Z. Gu, and L. Wang, "Distributed synthetic surrogate functional regression (D-SSFR): a scalable and robust framework for estimating spatially varying covariate effects in AI-augmented neuroimaging," *R&R at Annals of Applied Statistics*, 2026+. ([†]co-first author)

- W2. **Y. Long**, Z. Gu, G. Wang, and L. Wang, "Robust and scalable distributed learning for surface-based imaging regression with applications to neuroimaging," *In preparation*, 2026+.
- W3. **Y. Long**, Z. Gu, and L. Wang, "Synthetic surrogate image-on-scalar regression for robust brain imaging analysis," *In preparation*, 2026+.
- W4. **Y. Long**, D. Hanley, and D. Kepplinger, "Illuminant spectrum estimation and inference to study animal coloration from multispectral camera images," *In preparation*, 2026+.
- W5. **Y. Long**, C.E. Lee, Z. Li, "Conditionally uncorrelated components for replicated tensor time series via martingale difference divergence matrix," *In preparation*, 2026+.

PRESENTATIONS AND ACTIVITIES

Invited Talks

- | | |
|--|----------|
| T1. CFE-CMStatistics 2026 , <i>Virtual/Berlin, Germany</i> | Dec 2026 |
| Robust Estimation and Fast Robust Bootstrap Inference for Spatially Varying Coefficient Models | |
| T2. ICSA Applied Statistics Symposium 2026 , <i>Arlington, VA</i> | Jun 2026 |
| Distributed Synthetic Surrogate Functional Regression (D-SSFR) for AI-Augmented Neuroimaging | |
| T3. EcoSta 2025 , <i>Virtual/Tokyo, Japan</i> | Aug 2025 |
| Robust and Scalable Distributed Learning for Surface-Based Imaging Regression | |
| T4. ICSA Applied Statistics Symposium 2025 , <i>Storrs, CT</i> | Jun 2025 |
| Robust and Scalable Distributed Learning for Surface-Based Imaging Regression | |
| T5. 2025 Statistical Methods in Imaging Conference , <i>Houston, TX</i> | May 2025 |
| Robust Mean Signal Estimation and Inference for Imaging Data | |
| T6. StatConnect 2025 , <i>Fairfax, VA</i> | Mar 2025 |
| Robust Mean Signal Estimation and Inference for Imaging Data | |
| T7. ICORS meets DSSV 2024 , <i>Fairfax, VA</i> | Jul 2024 |
| Robust Learning and Inference for Mean Functions in Functional Data Analysis of Imaging Data | |

Contributed Talks

- | | |
|--|----------|
| T8. 2025 Joint Statistical Meetings , <i>Nashville, TN</i> | Aug 2025 |
| Robust and Scalable Distributed Learning for Surface-Based Imaging Regression | |
| T9. 2024 Joint Statistical Meetings , <i>Portland, OR</i> | Aug 2024 |
| Illuminant Spectrum Estimation to Study Animal Coloration from Multispectral Camera Images | |

Poster Presentations

- | | |
|---|------------|
| T10. GMU Department of Statistics 1st Statistics Showcase , <i>Fairfax, VA</i> | April 2026 |
| Synthetic Surrogate Functional Regression (SSFR) for AI-Augmented Neuroimaging | |
| T11. StatConnect@AI 2026 , <i>Washington, DC</i> | Mar 2026 |
| Synthetic Surrogate Functional Regression (SSFR) for AI-Augmented Neuroimaging | |

T12. GMU College of Engineering and Computing Innovation Week 2026 , <i>Fairfax, VA</i> Synthetic Surrogate Functional Regression (SSFR) for AI-Augmented Neuroimaging	Feb 2026
T13. The Past, Present and Future of Statistics in the Era of AI , <i>Washington, D.C.</i> Robust Mean Signal Estimation and Inference for Imaging Data	May 2025
T14. The Conference on Evolving Statistical Data Science , <i>Fairfax, VA</i> Accelerated Algorithms for Elastic Net S-Estimators	Mar 2023

HONORS & AWARDS

• Student Paper Honorable Mention , ICSA Applied Statistics Symposium 2026	2026
• Best Poster Presentation , GMU Department of Statistics 1st Statistics Showcase	2026
• Second Place in Student Poster Competition , StatConnect@AI 2026	2026
• Dissertation Completion Grant , George Mason University	2026
• R. Clifton Bailey Travel Award , Department of Statistics, George Mason University	2025
• Graduate Student Travel Fund , George Mason University	2024, 2025
• NSF Travel Award for ICSA , ICSA Applied Statistics Symposium 2025	2025
• Student Paper Runner-Up (Theory and Methods) , 2025 Statistical Methods in Imaging Conference	2025
• WSS Outstanding Graduate Student Award , Washington Statistical Society	2025
• Academic Excellence Scholarship in MS Statistics (one recipient annually) , Baruch College	2019
• Merit-Based Scholarship , Simon Business School, University of Rochester	2014

TEACHING EXPERIENCE

Instructor

• STAT 346: Probability for Engineers , George Mason University	Fall 2025
--	-----------

Teaching Assistant

• STAT 250: Introductory Statistics I , George Mason University	Spring 2021
• STA 2000: Business Statistics I , Baruch College	Fall 2017, Spring 2018
• STA 3154: Business Statistics II , Baruch College	Fall 2017, Fall 2018
• STA 9719: Foundations of Statistical Inference (Graduate) , Baruch College	Spring 2018

Guest Lectures

• STAT 663: Statistical Graphics and Data Visualization , George Mason University Introduction to Brain Imaging and Generative AI Imaging	Fall 2025
---	-----------

PROFESSIONAL EXPERIENCE

- Summer Associate (Data Scientist)**, Navy Federal Credit Union, *Vienna, VA* May 2024 – Aug 2024
- Enhanced credit card probability of default models for the CECL quantitative modeling team
- Quantitative Analyst Intern**, Truist Bank, *Charlotte, NC* Jun 2023 – Aug 2023
- Developed a machine learning framework with SAS and Python for suspicious transaction monitoring
- Graduate Assistant**, Statistical Consulting Laboratory, Baruch College, *New York, NY* Aug 2017 – May 2019
- Advised business school faculty and graduate students on data visualization and statistical software
- Quantitative Research Associate**, Terrapin Asset Management, LLC, *New York, NY* Oct 2015 – Dec 2016
- Performed empirical data analysis on hedge fund activism to validate and enhance a new hedge fund strategy

SERVICE TO THE PROFESSION

Conference Session Chair

- 2025 Joint Statistical Meetings**, *Nashville, TN* Aug 2025
Functional Regression for Complex Data: Accommodating Ordinal, Truncated, and Zero-Inflated Structures

Conference Committee

- IT and Website Committee Member**, ICSA Applied Statistics Symposium 2026, *Arlington, VA* Jun 2026

Journal Referee

- Journal of Applied Statistics: Environmental Statistics and Data Science (1 article)
- Journal of Computational and Graphical Statistics (1 article)
- Journal of Nonparametric Statistics (2 articles)
- Journal of Statistical Computation and Simulation (1 article)
- Statistical Analysis and Data Mining (2 articles)
- TEST (1 article)

Conference Volunteer

- ICSA Applied Statistics Symposium 2026**, *Arlington, VA* Jun 2026
- StatConnect 2025**, *Fairfax, VA* Mar 2025
- ICORS meets DSSV 2024**, *Fairfax, VA* Jul 2024
- IMS Meeting of New Researchers in Statistics and Probability**, *Fairfax, VA* Aug 2022
- SC21 (ACM/IEEE Supercomputing Conference)**, *Virtual/St. Louis, MO* Nov 2021

UNIVERSITY SERVICES

George Mason University

- President/Vice President**, Statistics Graduate Student Association (SGSA) Jun 2021 – May 2026
- PhD Representative**, Graduate and Professional Student Association (GAPSA) 2021 – 2023, 2025 – 2026